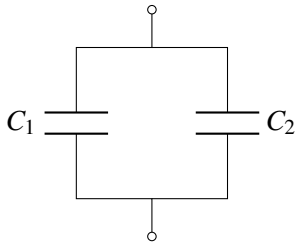

EECS 16A Designing Information Devices and Systems I

Discussion 10B

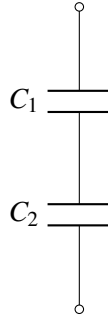
1. Practice: Series And Parallel Capacitors

Derive C_{eq} for the following circuits.

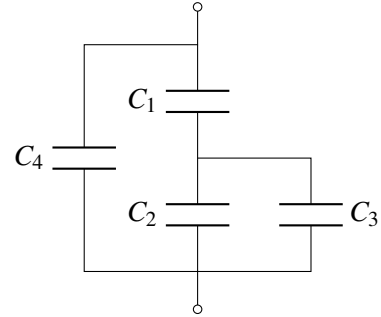
(a)



(b)

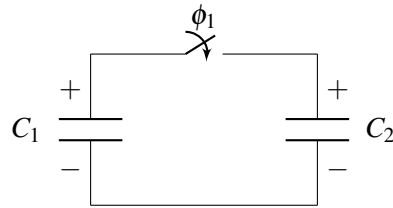


(c)

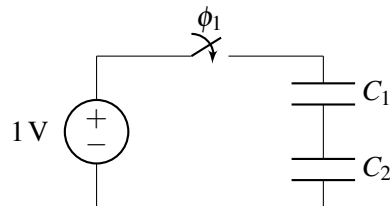


2. Capacitors and Charge Conservation

- (a) Consider the circuit below with $C_1 = C_2 = 1\ \mu\text{F}$ and an open switch. Suppose that C_1 is initially charged to $+1\ \text{V}$ and that C_2 is charged to $+2\ \text{V}$. How much charge is on C_1 and C_2 ?



- (b) Now the switch is closed (i.e. the capacitors are connected together.) What are the voltages across and the charges on C_1 and C_2 ?
- (c) Consider the following circuit with $C_1 = 1\ \mu\text{F}$ and $C_2 = 3\ \mu\text{F}$. Suppose that both capacitors are initially uncharged ($0\ \text{V}$).



What are the voltages across each capacitor after the switch is closed? What are the charges on each capacitor?

3. Bio-Molecule Detector [PRACTICE]

We've already seen how to build a bio-molecule detector where bio-molecules change the resistance between two electrodes. In this problem, we will explore a different scheme, where bio-molecules change the capacitance between two electrodes.

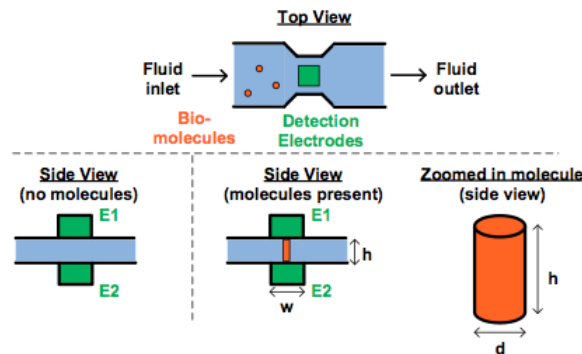


Figure 1: Bio-molecule detector.

As shown in Figure 1, the detector works by flowing a liquid that may or may not contain the biomolecules through a region in the device that has electrodes on the top and bottom of the liquid channel. The electrodes (E1/E2 in Figure 1) are chemically “functionalized” (using e.g. some appropriately designed antibodies), so that if the specific bio-molecule of interest is present in the fluid sample, one or more of the molecules will get physically trapped between the two electrodes (bottom right of Figure 1). After all of the fluid has been cleared out of the device (i.e., so that if there are bio-molecules present, there is only air in between the two electrodes E1/E2), we can then figure out whether or not one or more bio-molecules were trapped by measuring the capacitance between the two electrodes.

- If no bio-molecules are present, what should the capacitance between E1/E2 be? Assume that the electrodes are $10\mu\text{m}$ on each side and that the electrodes are 100nm apart. Leave your answer in terms of ϵ_0 .
- As shown in Figure 1, if each bio-molecule is a cylinder with diameter $d = 2\mu\text{m}$, height $h = 100\text{nm}$, and permittivity $\epsilon_b = 4\epsilon_0$, what would the capacitance between E1 and E2 be if only a single bio-molecule is trapped? Note that you can assume that the trapped molecule is exactly vertically oriented when it is trapped – i.e., the top and bottom faces of the molecule are both aligned with surfaces of the electrodes.
- Using the same numbers for d , h , and ϵ_b as in part (b), as a function of the number of trapped bio-molecules $N_{\text{molecules}}$, what is the capacitance between E1 and E2? (Note that you can assume that $N_{\text{molecules}}$ is small enough that all of the molecules fit within the electrode area and that all of the molecules are still trapped in exactly vertical orientation.)